Practicals

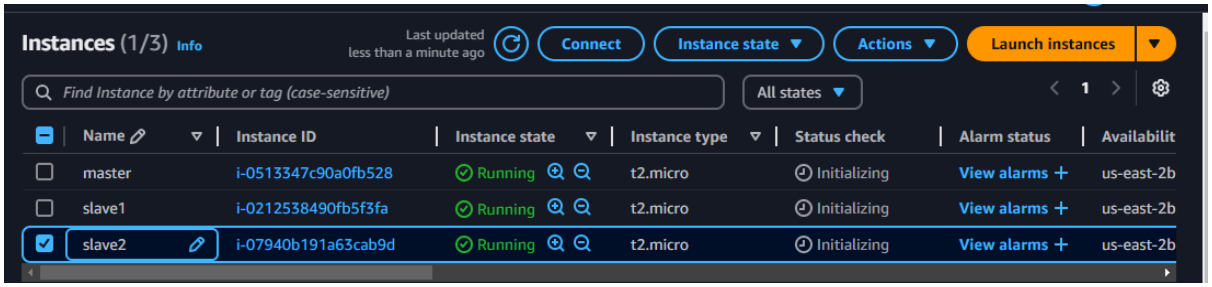
lab setup

launch 3 instance

1. install ansible on master
2. configure ssh access to ansible host
3. setting up ansible host and testing connection

one is master and both are client

launch 3 ubuntu server with normal configuration



create a sg with all traffic allowed in it and connect it to your server

one client and one master

>connect your master server

```
Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-31-209:~$
```

i-0513347c90a0fb528 (master)

>sudo apt update && sudo apt upgrade

```
ubuntu@ip-172-31-31-209:~$ sudo apt update
Hit:1 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 k
```

>sudo apt install software-properties-common

```
ubuntu@ip-172-31-31-209:~$ sudo apt install software-properties-common
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
software-properties-common is already the newest version (0.99.49.1).
software-properties-common set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ubuntu@ip-172-31-31-209:~$ sudo apt-add-repository  ppa:ansible/ansible
Repository: 'Types: deb
URIs: https://ppa.launchpadcontent.net/ansible/ansible/ubuntu/
```

> sudo add-apt-repository --yes --update ppa:ansible/ansible

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-30-40:~$  sudo add-apt-repository --yes --update ppa:ansible/ansible
Repository: 'Types: deb
URIs: https://ppa.launchpadcontent.net/ansible/ansible/ubuntu/
Suites: noble
Components: main
'
```

>sudo apt install ansible -y

```
ubuntu@ip-172-31-31-209:~$ sudo apt install ansible
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ansible-core python3-kerberos python3-nacl python3-ntlm-auth python3-p
  python3-xmltodict sshpass
```

>ansible –version

```
ubuntu@ip-172-31-19-45:~$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/ubuntu/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/ubuntu/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Nov  6 2024, 18:32:19) [GCC 13.2.0] (/usr/bin/python3)
  jinja version = 3.1.2
```

2. step

ssh access

now when you try to ssh your slave it will not happen

>cd .ssh >ls >ssh-keygen

```
ubuntu@ip-172-31-38-32:~$ cd .ssh
ubuntu@ip-172-31-38-32:~/.ssh$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/ubuntu/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ubuntu/.ssh/id_ed25519
Your public key has been saved in /home/ubuntu/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:JceUxsd0LdHo/cNxlpbZlGsiSmqAjB5ugy2m8u16Yk8 ubuntu@ip-172-31-38-32
The key's randomart image is:
```

>ls

>cat id_rsa.pub

```
ubuntu@ip-172-31-31-209:~/.ssh$ cat id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIApR6YSmVpHc+V4S4epUhwti2VnaNObt0BcM9W20dKRI ubuntu@ip-172-31-31-209
ubuntu@ip-172-31-31-209:~/.ssh$
```

on the both node server

cd .ssh copy that ssh.pub file here

sudo vi auth

```
ubuntu@ip-172-31-37-108:~$ cd .ssh
ubuntu@ip-172-31-37-108:~/.ssh$ sudo vi authorized_keys
ubuntu@ip-172-31-37-108:~/.ssh$
```

```
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQDK5sIcbPFKFC63fEwwOqLaO+9/IqBfILHQKTVnI0JXGFlIQUBuY721NwDHFMocYur5YiFhYQVkpiEHHih+saaGwyrlEtqrRx+BDhAluyJn4kC/z47/lmz
vF986KfMDQwf20h0th58sMSxSttoiwecYGqmjRucSZMllhqc7Ocfh0qRbFiUrP8dFGeDKLgdmxwk5Hs6sPJOwcwUmGvl/ZbVSANjdeuMbaQ6K47OX4WylI4mfcSFafShLKQ6C8HWLARM5OH9GYtDDRV2Z4W
gpgS8g3GbecbtlAlMOlq0wNMf+6Hd4i/gJb7b36qmKME/r0PCAaPD2FgIPBE3XEkEjulLn ansible
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIOXeMOZKpQlKnrcHQ37JbH3ISx1lQEKRMUAmMsBHnVaM ubuntu@ip-172-31-38-32
~
~
~
~
~
```

copy that file and paste it inside authorized_keys

now on the slave

>sudo apt update && sudo apt upgrade -y

>sudo apt-get install python3

```
ubuntu@ip-172-31-19-45:~$ sudo apt upgrade && sudo apt install python -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
```

```
ubuntu@ip-172-31-37-108:~/.ssh$ sudo apt-get install python3
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3 is already the newest version (3.12.3-0ubuntu2).
python3 set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.
ubuntu@ip-172-31-37-108:~/.ssh$
```

create a .pem file of your node server and add it in the control panel

>sudo vi demo.pem

pasted your key here from the window

>sudo chmod 400 demo.pem

```
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIOXeMOZKpQlKnrcHQ37JbH3ISx
ubuntu@ip-172-31-38-32:~/.ssh$ cd
ubuntu@ip-172-31-38-32:~$ sudo vi demo.pem
ubuntu@ip-172-31-38-32:~$ sudo chmod 400 demo.pem
ubuntu@ip-172-31-38-32:~$
```

now try to do ssh

>ssh ubuntu@ip

```
ubuntu@ip-172-31-38-32:~$ ssh ubuntu@16.171.31.125
The authenticity of host '16.171.31.125 (16.171.31.125)' can't be established.
ED25519 key fingerprint is SHA256:DPjV7hILX853twEGSQu3PplBUls9RYjQCFke6/gJUaE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '16.171.31.125' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-1021-aws x86_64)

 * Documentation:  https://help.ubuntu.com
```

to test your ansible

>ansible localhost -m ping

```
ubuntu@ip-172-31-19-45:~$ ansible  localhost -m ping
[WARNING]: No inventory was parsed, only implicit localhost is available
localhost | SUCCESS => {
    "changed": false,
    "ping": "pong"
}
```

>sudo chmod 766 ansible.cfg

now if you read its config file you won't me seeing much for that

>sudo ansible-config init --disable -t all > ansible.cfg

```
ubuntu@ip-172-31-38-32:/etc/ansible$ sudo chmod 766 ansible.cfg
ubuntu@ip-172-31-38-32:/etc/ansible$ sudo ansible-config init --disable -t all > ansible.cfg
ubuntu@ip-172-31-38-32:/etc/ansible$ ls
ansible.cfg  hosts  roles
ubuntu@ip-172-31-38-32:/etc/ansible$
```

create user and group

>vi /etc/ansible/host

add your slave ip here

[prod]

mantu ansible_ssh_host=51.20.85.255 ansible_user=ubuntu
ansible_ssh_private_key_file=/home/ubuntu/demo.pem

[test]

chantu ansible_ssh_host=16.171.31.125 ansible_user=ubuntu
ansible_ssh_private_key_file=/home/ubuntu/demo.pem

```
[prod]
mantu ansible_ssh_host=51.20.85.255  ansible_user=ubuntu ansible_ssh_private_key_file=/home/ubuntu/demo.pem

[test]
chantu ansible_ssh_host=16.171.31.125 ansible_user=ubuntu ansible_ssh_private_key_file=/home/ubuntu/demo.pem

## [webservers]
```

save it

>ansible-inventory –list

```
-r-------- 1 root root 1679 Feb  4 06:45 demo.pem
ubuntu@ip-172-31-38-32:~$ ansible-inventory --list
{
    "_meta": {
        "hostvars": {
            "chantu": {
                "ansible_ssh_host": "16.171.31.125",
                "ansible_ssh_private_key_file": "/home/ubuntu/demo.pem",
                "ansible_user": "ubuntu"
            },
            "mantu": {
                "ansible_ssh_host": "51.20.85.255",
                "ansible_ssh_private_key_file": "/home/ubuntu/demo.pem",
                "ansible_user": "ubuntu"
            }
        }
    },
    "all": {
        "children": [
            "ungrouped",
```

>sudo chown ubuntu:ubuntu /home/ubuntu/demo.pem

> chmod 400 /home/ubuntu/demo.pem

>ansible -m ping all

```
ubuntu@ip-172-31-30-40:/etc/ansible$ ansible -m ping all
[WARNING]: Platform linux on host mantu is using the discovered Python interpreter at /usr/bin/python3.12, but future installation of another Python
interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for r
information.
mantu | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.12"
    },
    "changed": false,
    "ping": "pong"
}
[WARNING]: Platform linux on host demo is using the discovered Python interpreter at /usr/bin/python3.12, but future installation of another Python
interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for r
information.
demo | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.12"
    },
    "changed": false,
    "ping": "pong"
}
```

if its success then its mean its done

>ansible -m ping production (group name)

>ansible mantu -m ping

```
ubuntu@ip-172-31-38-32:~$ ansible prod -m ping
[WARNING]: Platform linux on host mantu is using the discovered Python interpreter at /usr/bin/python3.12, but future installation of another Python
interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for mor
information.
mantu | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.12"
    },
    "changed": false,
    "ping": "pong"
}
ubuntu@ip-172-31-38-32:~$ ansible mantu -m ping
[WARNING]: Platform linux on host mantu is using the discovered Python interpreter at /usr/bin/python3.12, but future installation of another Python
interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for mor
information.
mantu | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.12"
    },
    "changed": false,
    "ping": "pong"
}
ubuntu@ip-172-31-38-32:~$
```

>ansible -m ping slave1 (server name)

```
ubuntu@ip-172-31-30-40:/etc/ansible$ ansible -m ping demo
[WARNING]: Platform linux on host demo is using the discovered Python interpr
interpreter could change the meaning of that path. See https://docs.ansible.c
information.
demo | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.12"
    },
    "changed": false,
    "ping": "pong"
}
ubuntu@ip-172-31-30-40:/etc/ansible$
```

if you dont have anyone so

```
ubuntu@ip-172-31-30-40:/etc/ansible$ ansible -m ping he
[WARNING]: Could not match supplied host pattern, ignoring: he
[WARNING]: No hosts matched, nothing to do
ubuntu@ip-172-31-30-40:/etc/ansible$
```

lets create a file for ping only

>mkdir playbooks

>cd playbooks

ansible-playbook --syntax-check ping.yml

```
ubuntu@ip-172-31-38-32:/etc/ansible/playbook$ ansible-playbook --syntax-check ping.yml

playbook: ping.yml
ubuntu@ip-172-31-38-32:/etc/ansible/playbook$
```